Yusuf Erdem

MSc AI Engineering Student at University of Passau

abdullahyusuferdem@gmail.com | 0163 7908221 | Passau, Germany | yusuferdem16.github.io

linkedin.com/in/yusuferdem16 | github.com/yusuferdem16

Summary

Early-career AI and software engineer with experience in multi-agent systems, machine learning, data science, ERP development, and GIS applications, currently pursuing an MSc in Artificial Intelligence Engineering at the University of Passau.

Education

University of Passau, MSc in Artificial Intelligence Engineering – Passau, Germany 2026 – present

Ankara Yildirim Beyazit University (AYBU), BSc in Computer Engineering – Ankara, Turkey 2022 – June 2026

- GPA: 3.13 / 4.0 (2.3 German scale)
- Relevant coursework: Machine Learning, Data Structures & Algorithms, Database Systems, Software Engineering, Computer Networks, NLP, Probability & Statistics, Neural Networks

Experience

AI Department Intern, Martur Fompak International – Bursa, Turkey Aug 2026 – Sept 2026

- Built a multi-agent AI system on Google ADK with a Gemini-based orchestrator coordinating knowledge-base search, parsing, and automated editing tools.
- Enabled full bidirectional cross-machine agent communication by integrating an independently built remote agent through the Agent-to-Agent (A2A) protocol.
- Designed a self-correcting Orchestrator/Editor/Reviewer review loop using ADK's LoopAgent and a reflection-style QA pattern.
- Delegated document edits to Groq-hosted LLMs to automate editing workflows.

IT Intern, SAP Logistics, Martur Fompak International – Bursa, Turkey July 2026 – Aug 2026

- Worked within a SAP-based Logistics IT team resolving factory workflow issues.
- Shadowed ABAP screen development to gain exposure to SAP development practices.
- Contributed to a Retrieval-Augmented Generation (RAG) chatbot for the Sales module. Converted internal documentation into a structured knowledge base.

Data Science Intern, Acun Medya Akademi – Remote Jan 2025 – Mar 2025

- Analyzed large-scale datasets using Python, SQL, Pandas, Matplotlib, and Seaborn.
- Applied data preprocessing, feature engineering, and statistical analysis methods to support business decisions.

Software Developer, Belsis – Ankara, Turkey Sept 2023 – Apr 2024

- Engineered custom Odoo ERP modules for inventory, sales, and BI reporting in Python, deployed to production supporting 50+ concurrent users.
- Managed PostgreSQL databases with pgAdmin and DBeaver.
- Used Git, GitLab, and Jira for team collaboration and workflow management.

Software Developer Intern, Belsis – Ankara, Turkey June 2023 – Sept 2023

- Designed a parcel-selection GIS application with OpenLayers adopted by multiple Turkish municipalities.
- Built the backend with C#, ASP.NET Core, PostgreSQL, and PostGIS.
- Built the frontend with Node.js, JavaScript, HTML, and CSS.

Projects

AI-Based Health Risk Scoring & Resource Allocation Jan 2025 – June 2026

- Co-developed a 3-person AI-based early warning and hospital-routing system for diabetes, hypertension, and stroke using exclusively non-laboratory, self-reportable patient data.
- Owned the hypertension risk model end-to-end and designed the risk stratification system.
- Integrated disease models with a hospital-routing optimizer and web interface.
- Awarded the TUBITAK 2209-A national undergraduate research grant.
- Presented at MUISC 2026 (Maltepe University).
- Technologies: Python, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn.

Skills

Programming Languages: Python, C#, JavaScript, SQL

Machine Learning and AI: TensorFlow, scikit-learn, PyTorch, FastText, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Agent-to-Agent (A2A) Protocol, LangGraph, MLOps, Reflection-style QA, ADK

Web and Backend Development: FastAPI, Flask, ASP.NET Core, Odoo, Node.js

Data Engineering and Analytics: PostgreSQL, PostGIS, ETL Pipelines, Feature Engineering, Pandas, Matplotlib, Seaborn

Tools and Collaboration: Docker, GitHub Actions, Git, GitLab, Jira, pgAdmin, DBeaver, Streamlit

Certifications

IELTS Academic 8.0 2025 – present

Complete A.I. & Machine Learning, Data Science Bootcamp Jan 2025 – present

Awards

Honorable Mention – Most Economical Project (Highway) UDHAM "Türkiye 2053"
Inter-University R&D Idea Competition, Ministry of Transport and Infrastructure

- Co-developed "Rotasyon," a congestion pricing and low-emission zone model to cut urban traffic and carbon emissions.
- Selected from 205 students across 62 universities and awarded a ₺50,000 prize.

Languages

Turkish: Native

English: C1

German: A2